

Application of non-isolated bidirectional DC–DC converters for renewable and sustainable energy systems: a review

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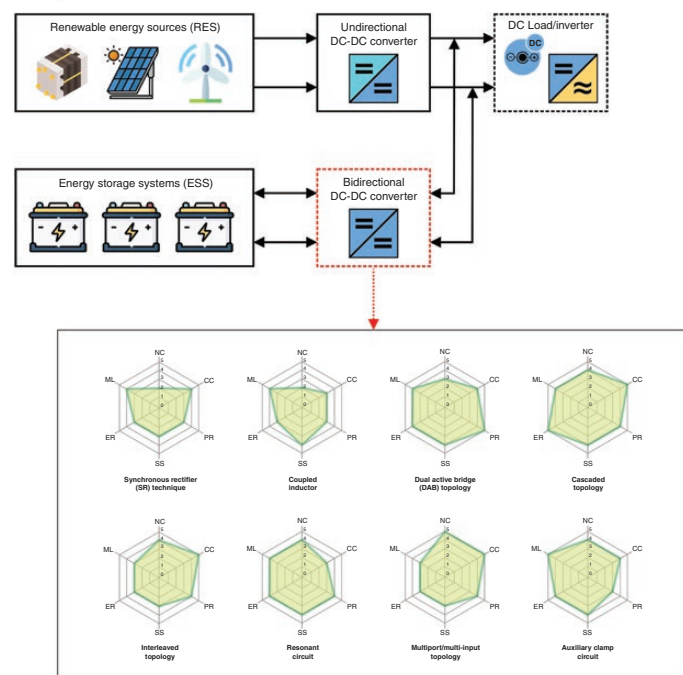
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Abstract

The primary challenge in renewable-energy utilization is an energy-storage system involving its power converter. The systems have to promise high efficiency, reliability and durability. Also, all of these can be realized at an economical cost. Buck and boost converters connected in parallel can convert power in both directions. It is the basic non-isolated bidirectional topology commonly used with energy-storage systems. The primary issue with the buck–boost non-isolated bidirectional converter is how to enhance its performance, so the modification involving this topology is still conducted. This paper examines 29 proposed converters from 30 research publications published in the last 10 years, the most recent of which focuses on modified non-isolated bidirectional converters based on the buck–boost topology. These are classified into eight modification schemes, which involve adding new components or circuits to the base topology. Each is evaluated against six parameters: the number of components, control complexity, power-rating applications, soft-switching ability, efficiency outcome and capacity to minimize losses. Moreover, each modified non-isolated bidirectional converter was compared from the renewable-energy-based power-generation-source perspective utilized. Based on these studies, researchers might think of ways to improve the buck–boost converter by changing it to make a new non-isolated bidirectional converter that can be used in systems that need it.

Graphical Abstract



Keywords: buck–boost converter; bidirectional converter; renewable energy; energy-storage systems; power converter

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Introduction

As power use increases, alternative energy must be developed to meet the requirements. Renewable energy sources (RES) have emerged as the best solution to sustainably provide clean energy while not polluting the environment [1]. In addition, the existence of a power-generation plant based on crude oil in the past made it scarce, as massive consumption contributed to environmental damage. This fact motivated policymakers and scientists to explore RES for power plants [2]. Photovoltaic (PV) systems, wind turbines (WTs), fuel cells (FCs), geothermal power plants, micro-hydro power plants and biogas power plants have been getting attention as clean and sustainable power plants. Nonetheless, the intermittent nature of RES, especially in power-plant-based PV systems and WTs, makes electricity production very challenging to meet the load demand [3]. Furthermore, low-voltage characteristics such as those generated by PV systems and FCs must be improved to meet the load-voltage requirement [4]. Therefore, to get high power conversion, advanced technology is needed, such as power electronic devices, control strategies and energy-storage systems (ESS).

PV systems convert light from solar energy into electricity through the PV effect [5]. The WT generates electricity by using the power of the wind to trigger an electric generator [6]. Meanwhile, an FC produces electricity from fuel via an electrochemical process [7]. The power generated can be distributed in three ways: grid-connected, freestanding or hybrid systems. Each power generation will be linked to other power plant systems in grid-connected systems. Stand-alone systems only generate power for loads close to the main power plant. Meanwhile, hybrid systems integrate both systems [8].

Freestanding systems are an effective source of electricity for remote locations that lack grid connectivity [9–11]. Consequently, storage systems should be used as storage devices because power generators [12], such as PV systems, cannot generate energy at night [13]. The bidirectional DC–DC converter, which interfaces the RES-based power-generation to storage systems (in charge mode) and sends storage power to the DC/AC inverter or DC load (in discharge mode), is one of the system components [14]. Furthermore, the bidirectional DC–DC converter is required to manage power processing during the transition from storage systems to load and overload [15]. The bidirectional DC–DC converter can be used in systems that require current to flow in both directions, depending on the mode of operation [16]. This is because it has certain features that make this possible.

Bidirectional DC–DC converters are available in two configurations: isolated topology and non-isolated topology [16]. Magnetic isolation separates the power transfer between the input and output in an isolated topology [17]. Non-isolated topology, on the other hand, does not involve isolation. The non-isolated conventional bidirectional DC–DC converter's primary and simplest topology is the buck–boost bidirectional converter [18]. When charging the storage systems, the converter runs in buck mode. When discharging electricity to meet the DC/AC inverter's voltage needs, it functions in boost mode.

However, the buck–boost bidirectional converter has some limitations, including not being suitable for high step-up or step-down due to hard switching, high duty-cycle operation, voltage-conversion ratio limitation and high voltage stress on the switching device [19]. As a result, researchers are constantly developing novel circuits that improve performance to address the disadvantages of the buck–boost bidirectional DC–DC converter design. Research is carried out due to the bidirectional

DC–DC converter's critical function in power generation, involving storage systems and the future exploitation of RES [20]. Therefore, numerous innovative bidirectional DC–DC converters have been presented to improve the classic architecture.

This study reviews the bidirectional DC–DC converter in renewable and sustainable energy systems. The review focuses on non-isolated bidirectional converter varieties of buck–boost topologies designed for use in RES-based power plants comprising PV, WT and FC generators. The second section describes the fundamental topology of the non-isolated DC–DC converter. The merits and disadvantages of each newly proposed topology based on the buck–boost converter will be discussed in Section 2. Section 3 compares each suggested converter and makes recommendations for future research on the development of the non-isolated DC–DC converter. Finally, Section 4 provides the article's conclusion.

1 Overview of non-isolated bidirectional DC–DC converter

The non-isolated bidirectional DC–DC converter has a fundamental topology of buck and boost coupled in anti-parallel, as shown in Fig. 1. This topology enables double-flow power transfer. As a result, programmable switching devices have replaced unidirectional semiconductors [21–24]. However, a high-frequency transformer-based isolated structure can be used to create a bidirectional DC–DC converter. A non-isolated topology is preferable because it can increase efficiency, allow a more compact design and is less expensive because it does not require a high-frequency transformer [25–27]. As a result, it is preferred when size and weight are the primary concerns [28–30]. In order to enhance the performance of the converter, the non-isolated bidirectional DC–DC converter has also been developed. It leads to a new architecture that includes single-ended primary inductor converter (SEPIC) converters [31–38], Zeta converters [39–44], Cuk converters [45–49] and Z-source converters [50–54], as shown in Figs 2–5.

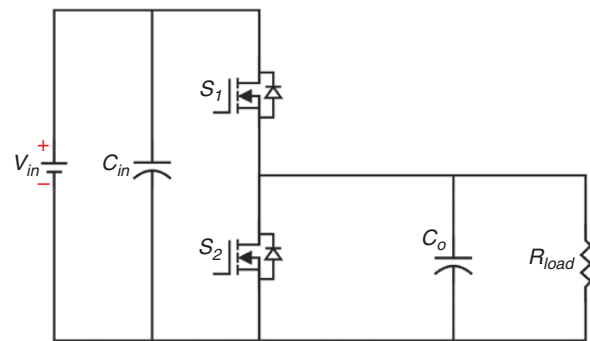


Fig. 1: Buck–boost converter.

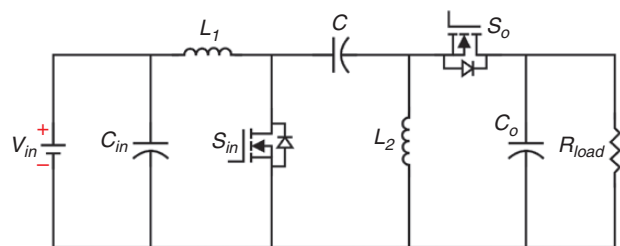


Fig. 2: SEPIC converter.

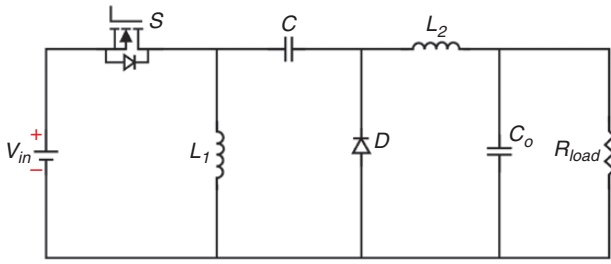


Fig. 3: Zeta converter.

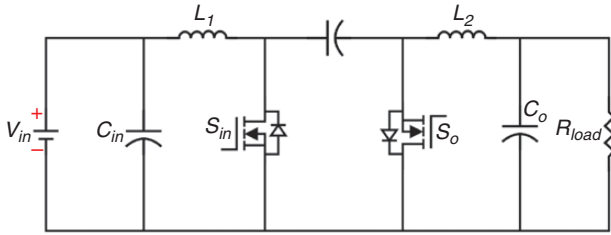


Fig. 4: Cuk converter.

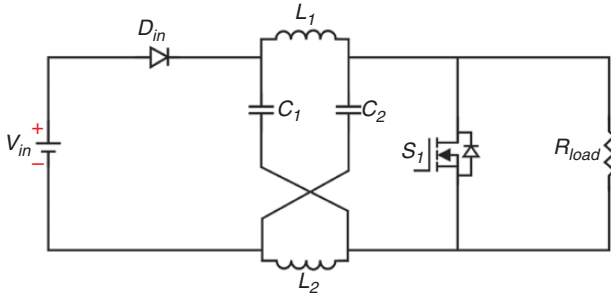


Fig. 5: Z-source converter.

The buck–boost bidirectional DC–DC converter is generally used to bridge the power source from RES-based power plants and storage systems, as illustrated in Fig. 6. The equivalent circuit is depicted in Fig. 7, which operates depending on the voltage source. The storage device is categorized as a low-voltage (L_v) side, while the high-voltage (H_v) side connects to the output voltage from the RES-based power plants, namely the DC bus. The buck mode flows from the H_v to the L_v side (charging), while the boost mode flows from the L_v side to the H_v side (discharging) [55]. So, this topology should use bidirectional power switches such as insulated gate bipolar transistors (IGBTs) or metal-oxide-semiconductor field-effect transistors (MOSFETs) that use a trigger pulse to allow power to flow in both directions.

According to what it does, the buck–boost bidirectional DC–DC converter has two modes of operation, which are the following.

1.1 Buck mode

In this mode, the converter is used to step down the output voltage to less than the source's input voltage, wherein its output voltage connects to the input terminal of the storage device. The diode of the switching device is vital in this topology. It allows the current to flow bi-directionally from the input voltage to the storage device and vice versa [56]. When the switch S_H is ON, S_L is kept OFF by activating the body diode, as illustrated in Fig. 8. The input current from the source rises and flows to the inductor L through S_H . Meanwhile, when S_L is OFF, the inductor current falls until the next duty cycle. In this

scheme, the charging process occurs due to energy storage in the inductor.

1.2 Boost mode

On the contrary, in this mode, the output voltage is stepped up and must be higher than the input voltage from the storage device (V_L). The power discharged flows to the load with switch S_L triggered and S_H is OFF with its body diode activated. The input current rises and goes to the inductor L and the switch S_L is ON. When S_L is OFF, the inductor current drops and will stop until the next cycle occurs. This process can also be conducted thanks to the switching device's diode. Fig. 9 shows the discharging/boost mode operation of the buck–boost bidirectional DC–DC converter.

There are five components to build a basic DC–DC bidirectional converter in a buck–boost topology. It consists of an inductor (L), two MOSFETs (S_H and S_L) and two capacitors (C_H and C_L). First, the duty cycle (D) must be determined. Equations 1 and 2 show how to do this:

$$D_{(\text{buck mode})} = \frac{V_L}{V_H} \quad (1)$$

$$D_{(\text{boost mode})} = \frac{V_H - V_L}{V_L} \quad (2)$$

The ripple in the inductor current (ΔI_L) should be defined to obtain the inductor value. It is commonly determined to be in the range of 20–40%. In addition, the switching frequency (f_{sw}) is determined to regulate the driving signal. Therefore, the inductor value can be calculated by:

$$L = \frac{(V_H - V_L) \times D}{\Delta I_L \times f_{sw}} \quad (3)$$

When determining the values of both capacitors, the equivalent series resistance (ESR) should be determined to minimize the output ripple. Equations 4 and 5 show the calculation of the high capacitor and the low capacitor, respectively, wherein ΔV_L and ΔV_H are the ripple voltage determined at 1% of each voltage:

$$C_L = \frac{(1 - D) \times V_L}{8L \times \Delta V_L \times f_{sw}^2} \quad (4)$$

$$C_H = \frac{I_{out} \times D}{\Delta V_H \times f_{sw}} \quad (5)$$

Furthermore, the estimated efficiency of the buck–boost bidirectional DC–DC converter can be obtained by considering each component applied. First, losses in MOSFETs, an inductor and capacitors are calculated based on each specification listed in the datasheet. Equations 6–8 show conduction loss, switching loss and reverse-recovery body diode loss in the high-side MOSFET (S_H). Meanwhile, losses on the low side of the MOSFET (S_L) are given in Equations 9–11, respectively:

$$P_{ON-H} = \left[I_{out}^2 + \frac{(I_{Lmax} - I_{Lmin})^2}{12} \right] \times R_{DS(on)-H} \times D_{(\text{buck mode})} \quad (6)$$

$$P_{S-H} = \frac{1}{2} \times V_H \times I_{out} \times (t_{r-H} + t_{f-H}) \times f_{sw} \quad (7)$$

$$P_{DIODE-H} = \frac{1}{2} \times V_H \times I_{RR-H} \times t_{RR-H} \times f_{sw} \quad (8)$$

$$P_{ON-L} = \left[I_{out}^2 + \frac{(I_{Lmax} - I_{Lmin})^2}{12} \right] \times R_{DS(on)-L} \times D_{(\text{boost mode})} \quad (9)$$

$$P_{S-L} = \frac{1}{2} \times V_L \times I_{out} \times (t_{r-L} + t_{f-L}) \times f_{sw} \quad (10)$$

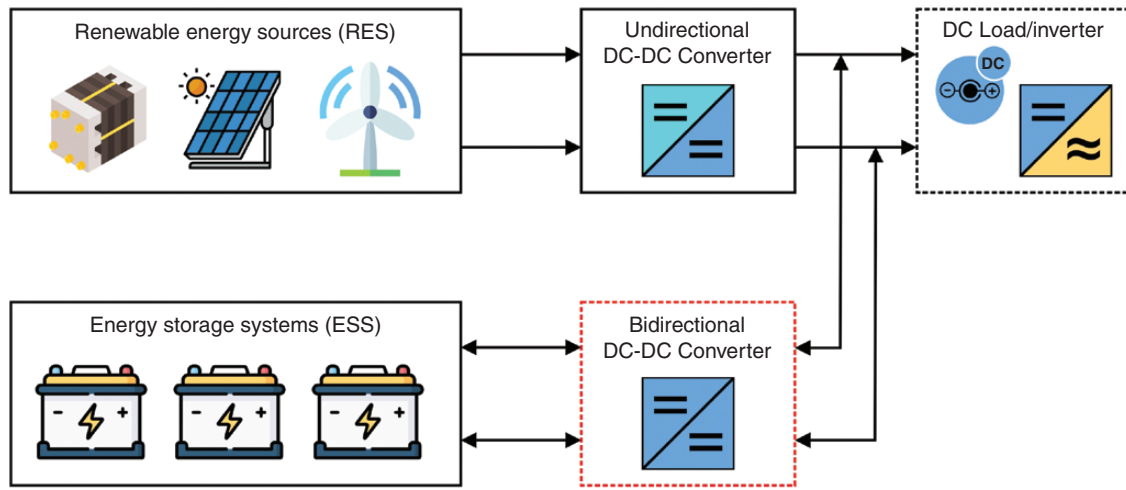


Fig. 6: Block diagram of RES-based power plants with energy-storage systems.

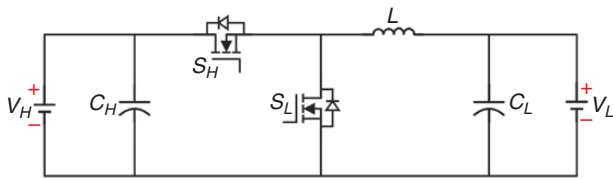


Fig. 7: Equivalent circuit of the buck-boost bidirectional DC-DC converter.

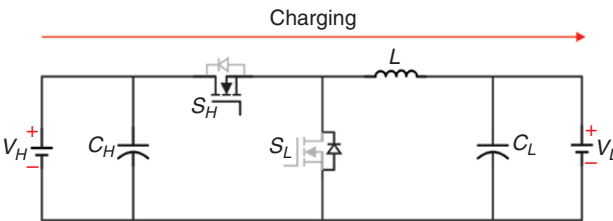


Fig. 8: The buck-boost bidirectional circuit in charging/buck mode.

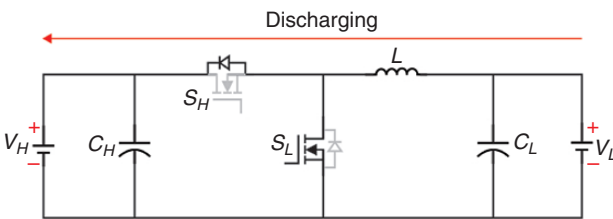


Fig. 9: The buck-boost bidirectional circuit in discharging/boost mode.

$$P_{DIODE-L} = \frac{1}{2} \times V_L \times I_{RR-L} \times t_{RR-L} \times f_{sw} \quad (11)$$

where P_{ON-H} represents the conduction loss in the high-side MOSFET, P_{S-H} represents the switching loss in the high-side MOSFET, $P_{DIODE-H}$ represents the reverse-recovery loss in the high-side MOSFET body diode, $R_{DS(on)-H}$ represents the on-resistance high-side MOSFET, t_{r-H} represents the rise-time high-side MOSFET, t_{f-H} represents the fall-time high-side MOSFET, I_{RR-H} represents the peak value of the body diode reverse-recovery current in the high-side MOSFET, t_{RR-H} represents the body diode reverse-recovery time in the high-side MOSFET, P_{ON-L} represents the conduction loss in the low-side MOSFET, P_{S-L} represents the switching loss in the low-side MOSFET, $P_{DIODE-L}$ represents the

reverse-recovery loss in the low-side MOSFET body diode, $R_{DS(on)-L}$ represents the on-resistance low-side MOSFET, t_{r-L} represents the rise-time high-side MOSFET, t_{f-L} represents the fall-time high-side MOSFET, I_{RR-L} represents the peak value of the body diode reverse-recovery current in the low-side MOSFET, t_{RR-L} represents the body diode reverse-recovery time in the low-side MOSFET, I_{Lmax} represents the maximum inductor current, I_{Lmin} represents the minimum inductor current and I_{out} represents the output current converter.

For each switching cycle, MOSFETs generate losses due to the output capacitance. The total losses of both the high-side MOSFET and the low-side MOSFET can be calculated using Equation 12:

$$P_{Coss} = \frac{1}{2} \times (C_{Oss-H} + C_{Oss-L}) \times (V_{H/L})^2 \times f_{sw} \quad (12)$$

where P_{Coss} represents the output capacitance loss in the MOSFET, C_{Oss-H} represents the high-side MOSFET output capacitance and C_{Oss-L} represents the low-side MOSFET output capacitance, while $V_{H/L}$ represents the high voltage or the low voltage depending on the operation mode of the buck-boost bidirectional DC-DC converter.

A dead-time scheme is needed to avoid a short circuit caused by both MOSFETs turning ON simultaneously, so the current spikes can be prevented when the inductor current flows continuously. Losses in dead time can be expressed using Equation 13:

$$P_D = V_D \times I_{out} \times (t_{Dr} + t_{Df}) \times f_{sw} \quad (13)$$

P_D represents the dead-time loss, V_D represents the forward direction voltage of the MOSFET body diode, t_{Dr} represents the dead time for rising and t_{Df} represents the dead time for falling.

The losses caused by charging the gate also occur in the MOSFETs. It depends on the high- or low-side MOSFET, which can be calculated using Equation 14:

$$P_G = V_{gs} \times (Q_{g-H} + Q_{g-L}) \times f_{sw} \quad (14)$$

where P_G represents the gate charge loss, V_{gs} represents the gate drive voltage, and Q_{g-H} and Q_{g-L} represent the gate charges of the high-side and low-side MOSFETs, respectively.

Furthermore, the losses caused by the operation of the integrated circuit (IC) and the conduction of the inductor used can be expressed using Equations 15 and 16, respectively. The inductor has two losses: the conduction loss due to internal resistance and the core loss caused by the magnetic element. But this article does not talk about the core loss in the inductor, which would have made it easier to do the maths:

$$P_{IC} = V_{H/L} \times I_{CC} \quad (15)$$

$$P_L = (I_{OUT})^2 \times DCR \quad (16)$$

I_{CC} represents the amount of current used by the IC and DCR represents the resistance to the direct current in the inductor.

Involving capacitors on the high side and low side also results in losses. It consists of leakage, dielectric and series resistance losses, in which these are simplified to ESR as a general loss. Equations 17 and 18 calculate out how much power is lost by the capacitor on the high side and the capacitor on the low side, respectively:

$$P_{C-H} = \left(I_{OUT} \times \frac{\sqrt{(V_H - V_L) \times V_L}}{V_H} \right) \times ESR_{C-H} \quad (17)$$

$$P_{C-L} = \left(\frac{\Delta I_L}{2\sqrt{3}} \right)^2 \times ESR_{C-L} \quad (18)$$

where the high-side capacitor loss is indicated by P_{C-H} , the low-side capacitor loss is indicated by P_{C-L} and the equivalent series resistance in the high-side and low-side capacitors is indicated by ESR_{C-H} and ESR_{C-L} , respectively. Therefore, the total loss P_{TOTAL} can be expressed as follows:

$$P_{TOTAL} = P_{ON-H} + P_{S-H} + P_{DIODE-H} + P_{ON-L} + P_{S-L} + P_{DIODE-L} + P_{COSS} + P_D + P_G + P_{IC} + P_L + P_{C-H} + P_{C-L} \quad (19)$$

Finally, the estimated efficiency (η) of the buck–boost bidirectional DC–DC converter can be calculated using Equation 19, in which P_{OUT} and P_{IN} are output power and input power, respectively:

$$\eta = \frac{P_{OUT} - P_{TOTAL}}{P_{IN}} \times 100 \quad (20)$$

2 Recent research of non-isolated bidirectional DC–DC converter

The development of the non-isolated bidirectional DC–DC converter almost does not involve a single technique. It can be combined with several techniques to produce high performance. However, this paper attempts to explain the closest method used by researchers to modify the conventional bidirectional DC–DC converter suitable for ESS in RES-based power plants. The focus of the modification technique is divided into eight, including the synchronous rectifier (SR) technique, coupled inductor (CI), dual-active-bridge (DAB) topology, cascaded topology, interleaved topology, resonant circuits, multiport/multi-input topology and auxiliary clamp circuit. These are based on applying additional components/circuits to the buck–boost topology.

2.1 SR technique

The SR technique is fundamental to yielding bidirectional topology. It replaces the diode in a unidirectional topology with an active switching device such as a MOSFET, IGBT or other controllable switching devices [57]. Therefore, the power flow can be carried in two directions. The benefit obtained is a reduction in conduction loss, which increases the efficiency result [58–61].

Additionally, this scheme has been added to improve the basic topology performance of the main switches. It has been carried out by Lin et al. [62], who proposed a buck–boost bidirectional converter employing additional switches in the main and synchronous in-series connection, respectively, so the number used is four switches, depicted in Fig. 10. As a result, this topology can obtain a wider voltage-conversion range than that of the conventional bidirectional converter. Also, the gain voltage in the step-up

mode can be limited by the fact that the components used are not perfect. This can be fixed.

A new modification topology presented by Sethuraman et al. [63] involves a single auxiliary inductor and capacitor, as shown in Fig. 11. In this case, it is arranged in a double boost circuit to increase the voltage gain. The switches used are more than the conventional bidirectional converter. This configuration can provide greater efficiency due to the current input flowing into two inductors, in which the motivation of this study is to enhance previous research [62]. By employing two inductors with different values, current flows are also different from each other and less than that in the converter [62]. Moreover, the inductor size can be reduced, minimizing the implementation cost and resulting in a promising dynamic response converter. On the contrary, one of the switches flows the total current from two inductors when in buck mode, wherein it is very high compared to the other switch current flows. Fortunately, by utilizing synchronous rectification, the switching losses in that switch are significantly decreased. In addition, the proposed converter can work for a wider range of the duty cycle, either in boost mode or buck mode.

2.2 CI

A CI is one of the components mostly used to modify the DC–DC converter. When the converter involves more than a single inductor, the CI becomes a solution to obtain several merits. It consists of voltage transformation, current ripple reduction, circuit impedance change and galvanic isolation, depending on the topology type. Additionally, by using this component, the leakage inductance can be minimized. The CI has the same characteristics in terms of the number-of-turns ratio. Furthermore, it can be found in special designs with a higher ratio, such as 1:1.5 to 1:50 and more [64].

For bidirectional topology, involving a CI is still viewed favourably, so many researchers conduct it. From [65], a proposed

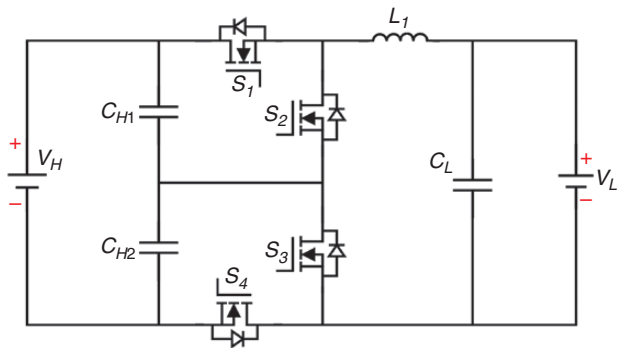


Fig. 10: Proposed converter based on the SR technique by Lin et al. [62].

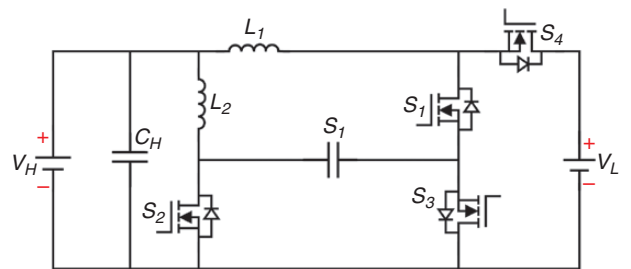


Fig. 11: Proposed converter based on the SR technique by Sethuraman et al. [63].

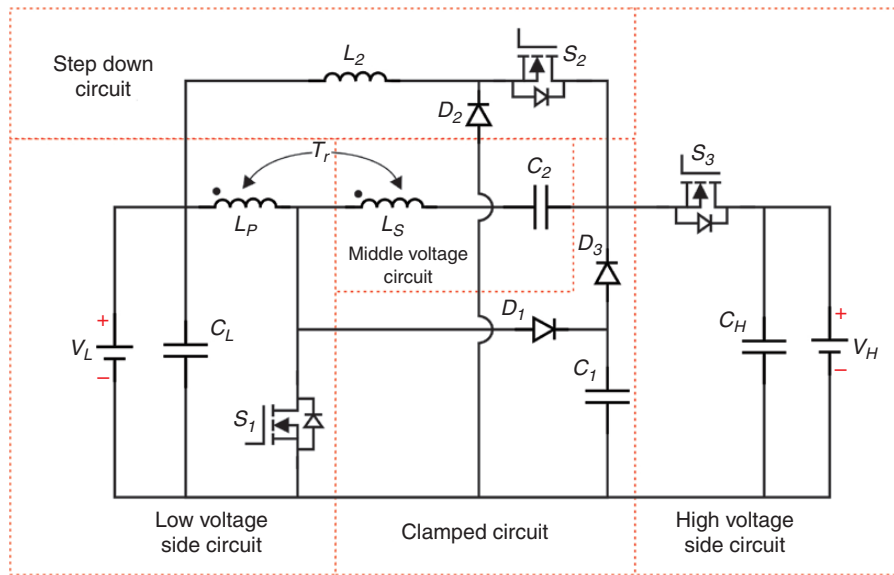


Fig. 12: Proposed converter based on the CI component by Wai et al. [65].

bidirectional DC-DC converter with a CI, which only uses three voltage-clamped power switches, is shown in Fig. 12. Also, an SR circuit and a soft-switching technique are added to reduce conduction losses and switching losses by applying devices with small $R_{DS(on)}$. A CI is useful in determining whether to charge or discharge energy, which means as a bidirectional magnetic switch. Leakage inductance restricts the changing voltage rate from the current rate. As a result, the timing of the current transition easily allows the switching devices to operate under the soft-switching conditions. Other benefits include the ability to solve saturation problems, unbalanced magnetizing current and reverse-recovery current in diodes. This leads to a high-efficiency bidirectional DC-DC converter with a simple feedback controller that uses the proportional integral (PI) algorithm.

In [66], a proposed bidirectional DC-DC converter was modified from the boost DC-DC converter in [67], as depicted in Fig. 13. The proposed converter utilizes a CI with the same winding ratio and can generate step-up/down voltage based on the series-parallel concept on both CI sides. When the converter operates in boost mode, both winding sides of the CI operate in parallel to charge and in a series connection to discharge. In contrast, in the buck mode, wherein both CI wind charges are in series mode and the discharges are in parallel, the proposed converter is suitable for generating high step-up and step-down voltages. Moreover, this converter can work in a bidirectional mode differently from the basic topology modified by replacing the diode with a MOSFET, which obtains a pulse-width-modulated (PWM) signal.

Fig. 14 shows another simple modification based on the buck-boost bidirectional converter proposed by Narasimharaju et al. [68], only replacing the existing inductor with a tapped inductor, also called a CI. This results in current stress, and voltage stress in diodes and switches can be significantly reduced. Therefore, both components do not require high specifications, reducing the cost. In addition, this converter involves low-priced control using voltage mode control in IC devices. Meanwhile, the proposed bidirectional topology of Saadatizadeh et al. [69] in Fig. 15 introduces the third winding at the CI and a capacitor to cancel the ripple of the input current. The proposed converter can increase the voltage-conversion result by adding the number of turn ratios on the CI second winding. Also, zero-voltage switching (ZVS) can

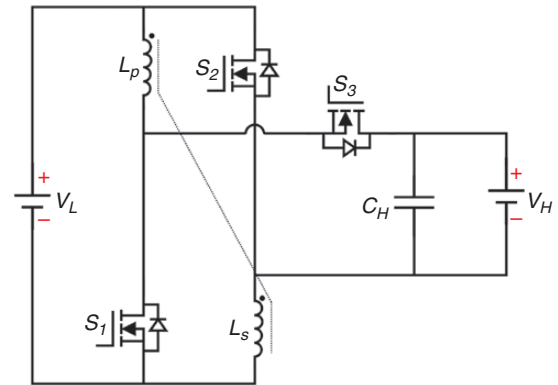


Fig. 13: Proposed converter based on the CI component by Yang and Liang [66].

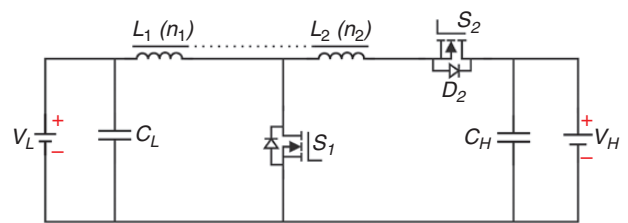


Fig. 14: Proposed converter based on the CI component by Narasimharaju et al. [68].

be done with just two MOSFETs across the whole power range by changing the value of the auxiliary inductor.

The bidirectional modification converter presented in [70] utilizes an intermediate storage capacitor combined with a CI, as shown in Fig. 16. The proposed converter also applies a clamped circuit followed by a multiphase scheme and a capacitor output. As a result, the proposed converter generates advantages such as high efficiency and low ripple. In addition, it offers the ability to operate the converter phase depending on the operation mode. The proposed converter operates in a two-phase configuration when boosting voltage, but only a single-phase in buck mode. Therefore, the proposed converter can achieve high voltage gain without an extreme duty cycle in the step-up mode, so the total

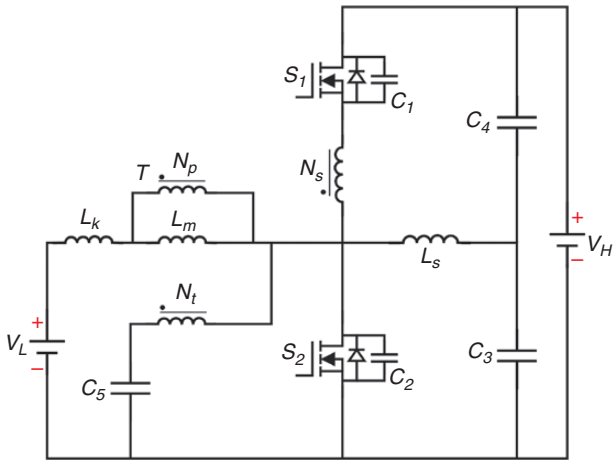


Fig. 15: Proposed converter based on the CI component by Saadatizadeh et al. [69].

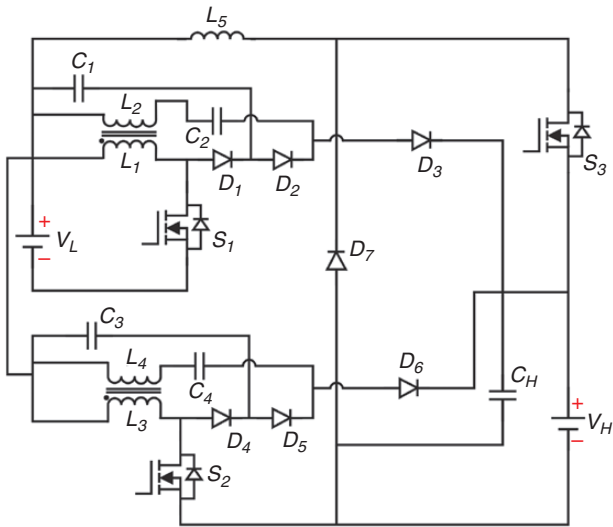


Fig. 16: Proposed converter based on the CI component by Hussain et al. [70].

voltage stress in the switching devices can be minimized because two switches are simultaneous. Moreover, employing a multi-phase configuration with the interleaving technique can handle ripple cancellation.

2.3 DAB topology

Due to its features [71], the DAB topology has received a lot of attention: ZVS characteristics, high power density with its handling capability, the ability to design in cascaded or modular circuits that provide a higher voltage and power due to galvanic isolation, and the flexibility to determine in non-isolated or isolated configurations [72–75]. Therefore, DAB topology research has been ongoing [76], on either the control systems or the circuit converter itself, to result in better performance. The challenge of applying the DAB topology is to regulate power and voltage with high efficiency. When the voltage between the two sides of the DAB topology is mismatched, negative effects occur such as high power losses during the turn-off switch period, failure to achieve soft-switching conditions and a high circulating current during converter operation [77]. So, new research combines the DAB topology with other circuit configurations to produce a new topology.

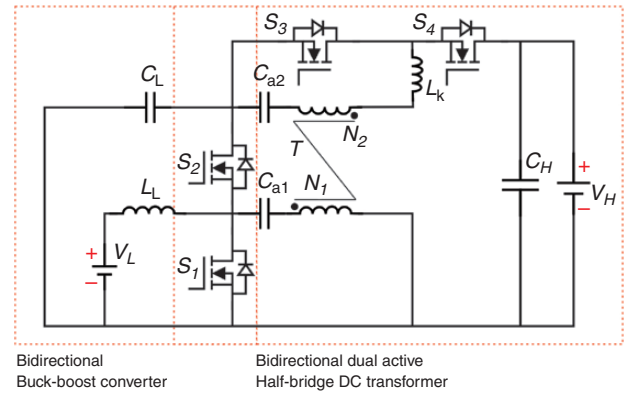


Fig. 17: Proposed converter based on DAB topology by Wu et al. [78].

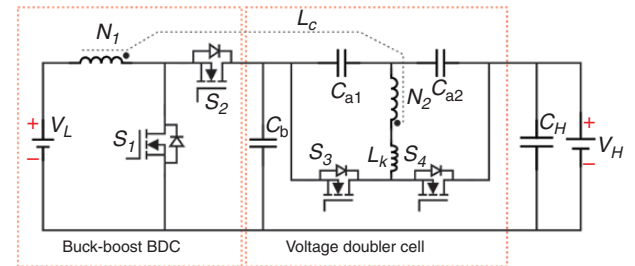


Fig. 18: Proposed converter based on DAB topology by Wu et al. [79].

The novel bidirectional DC–DC converter, which integrates two topologies, has been proposed by Wu et al. [78], consisting of a conventional buck–boost converter and a dual active half-bridge (DAHB) circuit, as shown in Fig. 17. It is carried out due to the conventional buck–boost being inefficient for high-voltage applications. Therefore, DAB is added by utilizing a DC transformer, with a CI also mentioned. The proposed configuration connects DAB in series, so the voltage stress through switches can be reduced. In addition, employing a DC transformer offers a high ratio of increasing and decreasing voltage conversion and a wider range. For the controller used, phase shift is applied to determine the duty cycles in each switch to match the voltage between the DC transformer's sides and regulate the power flow in the converter. So, all switches can get to ZVS and reduce conduction losses while the converter is working.

Furthermore, by adding a voltage doubler cell, as shown in Fig. 18, the topology modified in [79] functions the CI as an inductor filter in bidirectional buck–boost and as a DC transformer in the DAHB topology. The CI secondary winding is placed in the voltage doubler cell to achieve a higher bidirectional buck–boost voltage-conversion ratio. The PWM control generates duty cycles to regulate voltage matching in the bidirectional DAHB buck–boost. The voltage between the two sides is always matched and achieves soft-switching conditions that involve reducing conduction losses. Also, PWM phase-shift control can be used to control the flow of power in the proposed converter.

2.4 Cascaded topology

A cascading topology is used to achieve a larger conversion ratio. It typically combines more than a single converter topology into a series connection. The bidirectional buck–boost converter is called the cascaded buck–boost (CBB) topology, which can step up or step down the voltage in two modes. This topology is a four-quadrant converter and is viewed as the most flexible structure. However, several drawbacks still exist, such as the increased

control complexity due to increased power switches and conduction losses due to reverse recovery in body switches. On the other hand, compared to cascaded half buck-boost CHB, CBB has higher performance in terms of efficiency as it only uses a single inductor [80]. Therefore, this topology is still being developed to obtain better performance.

The simple modification carried out by Lee and Yun [81] is depicted in Fig. 19, adding a capacitor in series connection as an interface between the two CBB converters. Consequently, the ripple reduction can be achieved because the auxiliary capacitor also functions in the filter by providing a bypass track for the output current. Research by Teng et al. [82, 83] offered CBB modification in an interleaved configuration, which used the components more, as shown in Fig. 20. The operation mode can be carried out by various schemes: single, interleaved and cascading converters. It creates a more complete voltage-conversion ratio that can be used to connect the source voltage to the storage device and load.

2.5 Interleaved topology

Splitting the power transfer into multiple phases offers some advantages over a single phase, in which it can increase efficiency and decrease conduction loss. This configuration is also known as interleaved topology or multiphase topology [84], which operates with a given phase shift for each switching device. As a result, ripple cancellation can be achieved and the filter used can be minimized to avoid a slow dynamic response [85]. This topology has been commonly applied to bidirectional schemes. In [86], Pham et al. investigated three topologies of the bidirectional type, consisting of buck-boost, split buck-boost and interleaved dual-phase buck-boost, to show each performance. Based on the results of the tests, the interleaved configuration shown in Fig. 21 is the most effective.

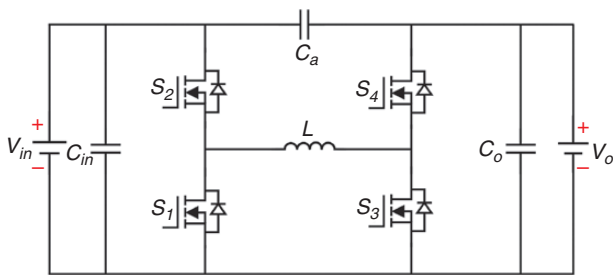


Fig. 19: Proposed converter based on the cascaded topology of Lee and Yun [81].

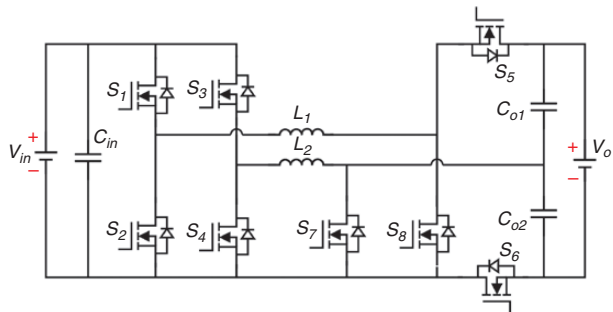


Fig. 20: Proposed converter based on cascaded topology by Teng et al. [82, 83].

A new structure based on the interleaved topology, as shown in Fig. 22, was introduced by Simões et al. [87] in a floating configuration mentioned as the bidirectional floating interleaved buck-boost converter (FIBBC). This topology consists of four phases and can achieve low input current ripple, high voltage gain and high efficiency in reducing the capacitor as a filter and inductor volumes compared to the conventional topology. The FIBBC topology is dedicated to PV systems with ESS. Therefore, it uses a control system that provides stand-alone and on-grid features.

Furthermore, the modification topology proposed by Liu et al. [88] involves the interleaved buck-boost converter and CLLC resonant converter, as shown in Fig. 23. The power flow in the charging mode places the CLLC topology in the first stage and an interleaved SR buck-boost converter in the second stage. Meanwhile, both converters are placed in the reverse stage in the discharging mode. This proposed circuit handles high-frequency switching operations and avoids phase error in the interleaving process. In addition, the previous research results were better when the proposed converter was run in triangular conduction mode (TCM) [89].

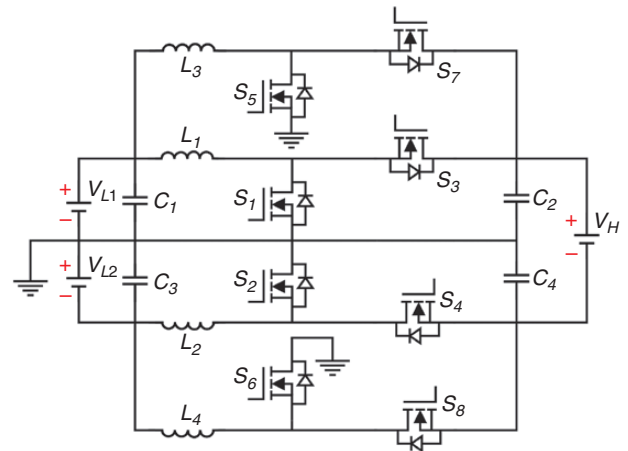


Fig. 21: Proposed converter based on the interleaved topology of Pham et al. [86].

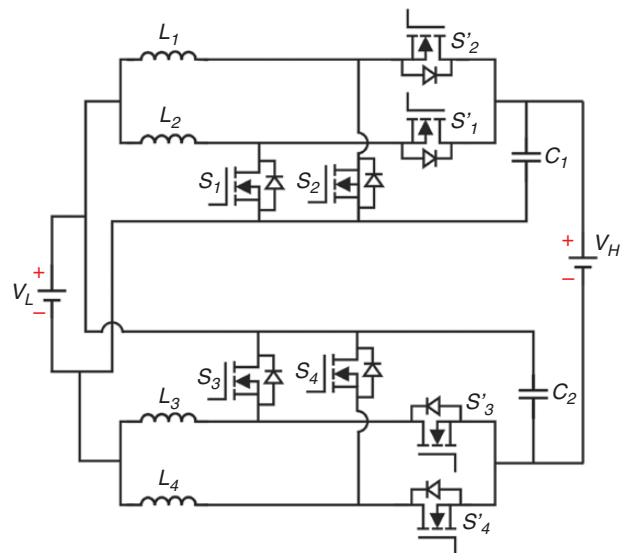


Fig. 22: Proposed converter based on the interleaved topology of Simões et al. [87].

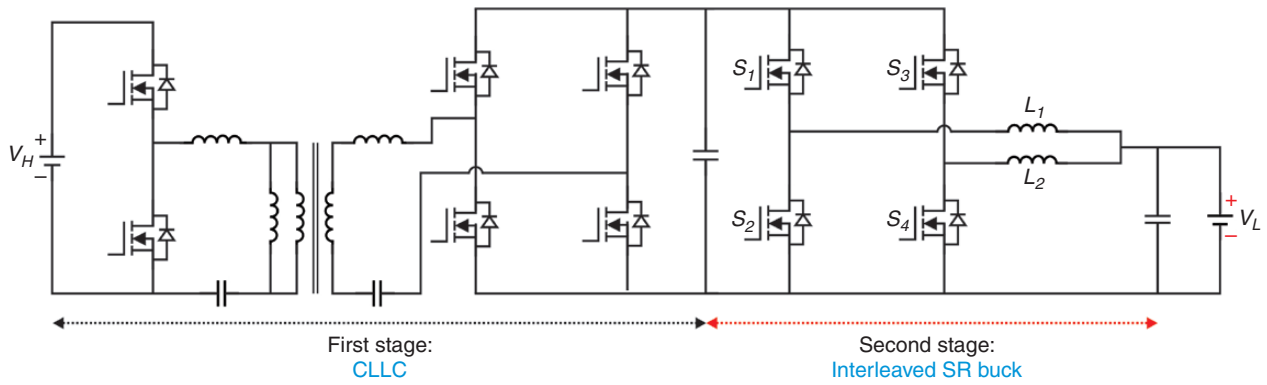


Fig. 23: Proposed converter based on the interleaved topology of Liu et al. [88].

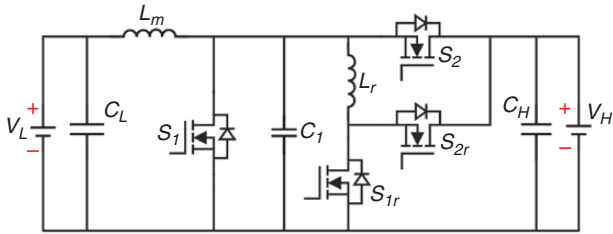


Fig. 24: Proposed converter based on the resonant circuit of Chao and Huang [94].

Although the interleaved topology has several advantages, several aspects should be considered when implementing it. This structure is constructed by two or more single converters in parallel connection and it means the number of components is greater than that used in a single topology, which implicates the size, weight, cost and control complexity itself. Also, one feature of the interleaved topology to obtain ripple cancellation is susceptible to non-ideal conditions, which create unbalanced current flow in each phase. It can be caused by the inductor, asymmetric layout and other non-ideal components [90]. Therefore, the control strategy should be good to avoid these problems [91] and modulation of the variable switching frequency has become a popular way to get the benefits of the interleaved topology.

2.6 Resonant circuit

The resonant circuit is applied for better performance by decreasing losses during the switching period [92]. It consists of at least one single inductor and capacitor in series or parallel connections, also mentioned as a resonant tank [93]. In addition, the resonant circuit can be developed by integrating auxiliary switches or combined with basic topology to produce promising performance. Like the modification carried out by Chao and Huang [94], a resonant branch with simple control for PV applications has been proposed, shown in Fig. 24, offering soft-switching conditions while maintaining primary switching devices in the ZVS area during the proposed converter operation. As a result, it is implicated in improving the converter efficiency.

Furthermore, the modification of the buck–boost bidirectional converter involving soft-switching cells has been proposed by research [95], as shown in Fig. 25. The soft-switching cell is an LC series-resonant circuit inserted between the high-input side (buck) and low-output side (boost) to optimize the storage device charge and discharge system. This configuration means that ZVS in the switches can be realized to avoid losses during converter operation. The same method is also adopted by Mohammadi and

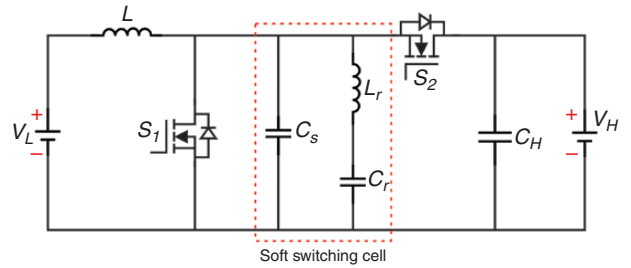


Fig. 25: Proposed converter based on resonant circuit by Jung et al. [95].

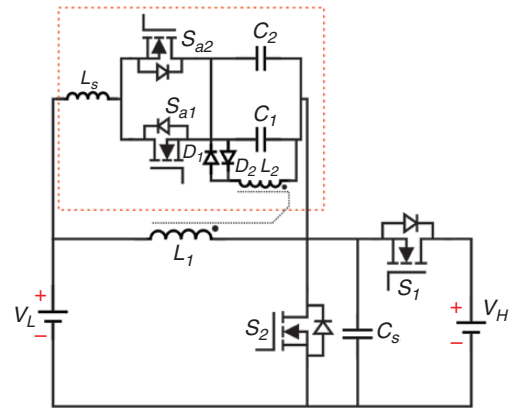


Fig. 26: Proposed converter based on the resonant circuit by Mohammadi and Farzanehfar [96].

Farzanehfar [96], which utilizes LC with additional switches in the buck–boost bidirectional converter in Fig. 26. These auxiliary circuits are connected in parallel with the primary inductor in the buck–boost converter. It is conducted to extend ZVS operation ranges, so the proposed circuit is more acceptable. Also, the voltage stress on the main switches can be reduced and the voltage stress on the auxiliary switches is less than that on the main switches.

Ashique and Salam [97] proposed a new strategy called zero-voltage zero-current switching (ZVZCS) to ensure that the soft-switching condition is obtained. It is realized by an auxiliary resonant network consisting of two switches, an inductor, a capacitor and a diode for the buck–boost converter, as shown in Fig. 27. Therefore, the modification circuit causes the utilization of two inductors that work as inductive snubbers and support achieving zero-current switching. As a result, the main switch can operate over a wide range of operating input-voltage, switching-frequency

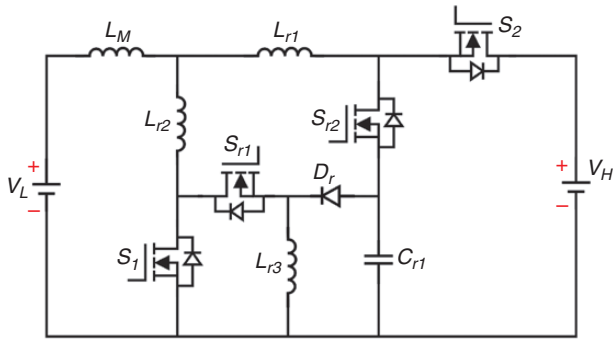


Fig. 27: Proposed converter based on the resonant circuit of Ashique and Salam [97].

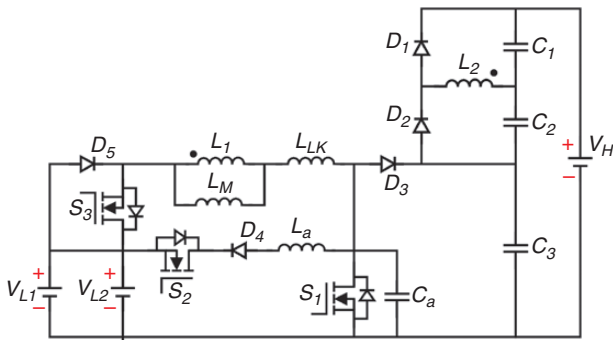


Fig. 28: Proposed converter based on multi-input topology by Faraji et al. [100].

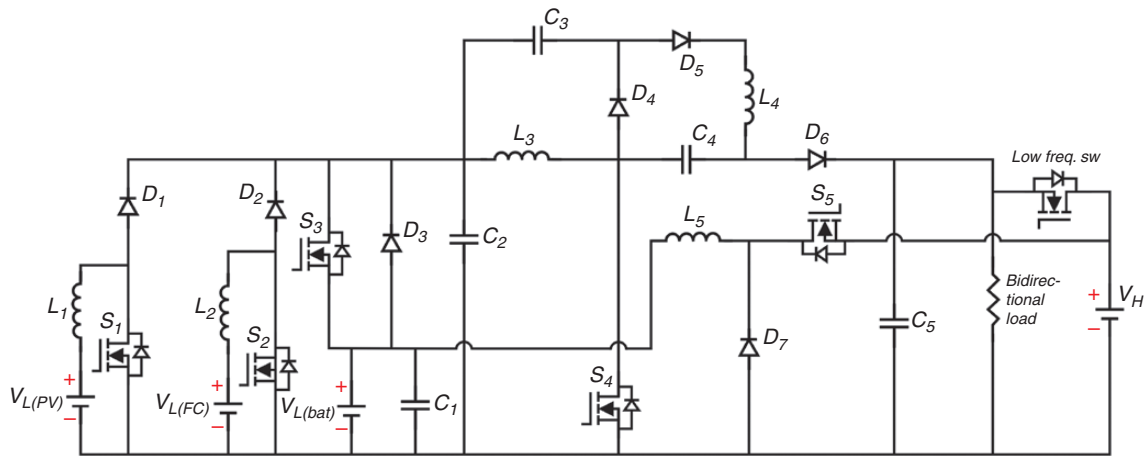


Fig. 29: Proposed converter based on multi-input topology of Ghavidel et al. [101].

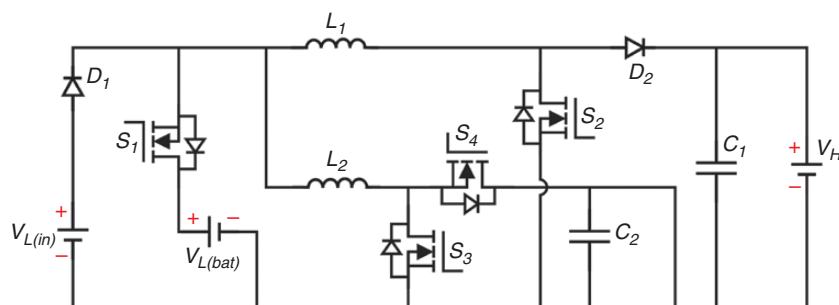


Fig. 30: Proposed converter based on multi-input topology by Khan et al. [102].

and load-current variations, promising to produce a high-efficiency converter over a wide operating range. Furthermore, the more complex modification of the bidirectional converter with the resonant circuit is provided by Uno and Sugiyama [98], involving a switched-capacitor circuit in a multiport converter. In addition, Rathore et al. [99] combine buck-boost converters with LCL resonant and voltage doubler circuits into a half-bridge configuration based on bidirectional converters.

2.7 Multiport/multi-input topology

Receiving energy from different sources such as PV cells, FCs and WT's at different voltages and power levels and then converting that energy into regulated voltages has created significant challenges in the design of power-conversion systems. Therefore, a DC-DC converter with a multiport/multi-input topology is designed to meet the system requirement. In some applications, such as hybrid electric vehicles or PV power systems, wherein isolation is not required, a non-isolated DC-DC converter is more suitable. It is chosen because of the simplicity, control complexity and economic cost of the component. Therefore, the study of multi-input converters is still considered by many researchers around the world. A new non-isolated DC-DC high step-up converter with a multi-input topology used to interface a hybrid power source is proposed in [100], as shown in Fig. 28. This converter can increase the number of inputs and transfer power from a hybrid source such as a PV system and FC to hybrid ESS such as batteries and supercapacitors (SCs) through appropriate power lines. This converter is designed to use only three switches for two input sources and is capable of charging and discharging energy-storage devices. It also reduces the complexity of the design and implementation costs.

Fig. 29 shows the proposed converter of Ghavidel *et al.* [101] consisting of two unidirectional input power ports and two bidirectional ports for storage elements in an integrated structure. The proposed converter has several input sources with different voltage levels set over a medium voltage range. The suggested structure uses a boost-type converter, a multi-input and single-output converter, and renewable sources. The recommended structures are designed for flexible energy systems such as smart cities, smart homes and electric vehicles. Meanwhile, in the study in [102], the author proposed a multi-input multi-output bidirectional non-isolated DC–DC converter, as illustrated in Fig. 30. The designed converter provides different voltage levels at the output with inter-source power-sharing and bidirectional power-flow operation capability. Moreover, the proposed converter attained high efficiency and is more beneficial for low-power applications.

A modified non-isolated bidirectional DC–DC converter with multiport/multi-input has been proposed in [103], as shown in Fig.

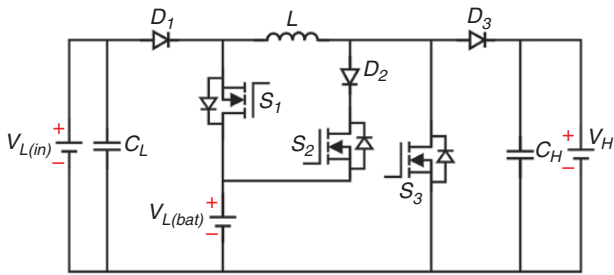


Fig. 31: Proposed converter based on multi-input topology by Mohapatra and Mohapatro [103].

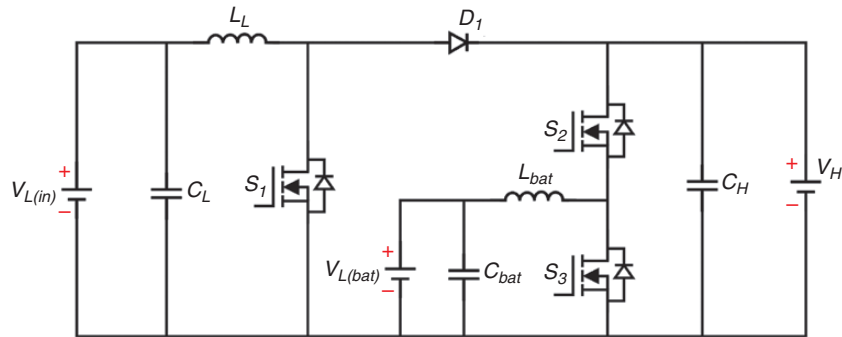


Fig. 32: Proposed converter based on multi-input topology by Wu and Lin [104].

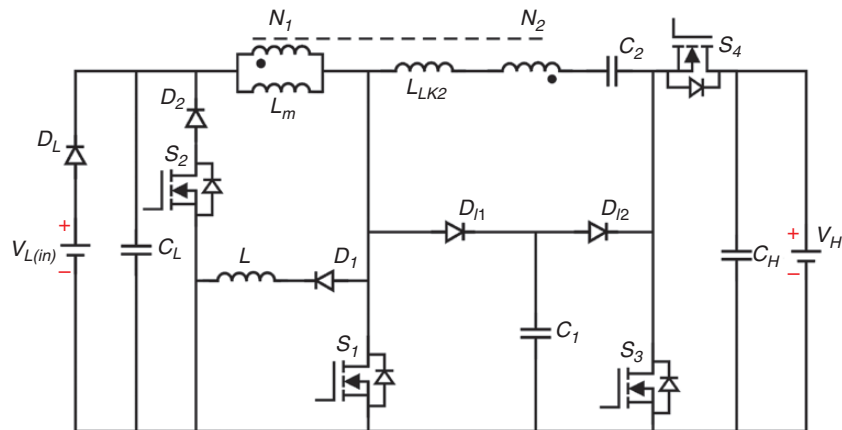


Fig. 33: Proposed converter based on multi-input topology of Wu and Hsiao [105].

31. For a 48-V microgrid, the converter can combine multiple sources, such as a PV source, WT, FC and a battery as a storage device. The proposed converter has more minor components, which makes the structure more compact. It has three ports, which contain only three controlled switches and one inductor. In [104], a three-port bidirectional non-isolated DC–DC converter for PV-based electric scooter charging stations is depicted in Fig. 32. The proposed converter combines an improved modified boost converter and a bidirectional buck–boost converter. The proposed converter reduces the number of components and overall cost by using a switch internal diode. Moreover, it is developed in [105], focusing on PV systems by combining the proposed converter with a CI and clamp capacitor to avoid voltage spikes in the main switches, as shown in Fig. 33. Meanwhile, [106] offers a non-isolated bidirectional DC–DC topology with a simple multiple input (MI), which can operate in boost, buck or buck–boost modes, as illustrated in Fig. 34. This modification promises to conduct the power flow in a bidirectional scheme. Therefore, the configuration involves a battery pack to store any excess energy from the input source that can be used to supply the load when needed. The main goal of this work is to come up with a way to simplify the buck–boost bidirectional converter's multi-input topology.

2.8 Auxiliary clamp circuit

On a non-isolated DC–DC converter, the auxiliary clamp circuit is often made up of an auxiliary switch, an auxiliary inductor and a clamp capacitor [107]. It is generally used to achieve a soft-switching condition of the converter to reduce switching losses when high-frequency switching is applied. Like other described soft-switching techniques, this strategy utilizes additional components to modify the basic topology, such as buck–boost circuits

based on bidirectional converters. The ZVS condition in the main switches is achieved by using an active auxiliary circuit consisting of auxiliary switches, diodes, an inductor and a capacitor, which has also been done by Yang and Do [108], as depicted in Fig. 35. It is possible to produce high efficiency by reducing switching losses. In addition, there are much fewer conduction losses in the main switches and auxiliary circuits, which makes the whole system more efficient.

Furthermore, Fig. 36 shows a novel soft-switching technique implemented in a non-isolated bidirectional DC–DC converter by using an active clamped auxiliary switch to achieve the inductor's continuous current mode (CCM) and operating at a constant frequency to turn ON and turn OFF the switches in [109]. The proposed converter uses only one auxiliary switch, two inductors and one capacitor, giving the converter a straightforward structure wherein the procedure of soft switching is obtained by ZVS in the main switch, while the auxiliary switch works on zero-current switching (ZCS). Also, by applying one more inductor, the proposed converter offers the active snubber configuration. It means that the reverse-recovery problem caused by the diode inside the switches can be fixed, which can help reduce conduction losses during operation mode.

In [110], the ZVS condition is secured for both the boost and buck operation modes of the converter operating the duty cycle or the load condition is achieved using a simple auxiliary circuit consisting of an auxiliary switch, an auxiliary inductor and a clamp capacitor, as shown in Fig. 37. The proposed topology does not need complex variable-frequency control methods to control its converter. In addition, the interleaved configuration can operate with the conventional fixed-frequency PWM control procedure. This study's primary focus is on how to simplify its control strategy. Meanwhile, the auxiliary clamp circuits in [111], as shown in Fig. 38, are dedicated to decreasing voltage spikes that cause CIs. It is a simple method that has been shown to prevent voltage stress and could improve the efficiency of the converter.

3 Discussion and comparison of modified non-isolated bidirectional DC–DC converter

Power transfer from power generation from RES into storage systems is required to result in a highly efficient conversion. The appropriate bidirectional DC–DC converter should be involved to realize it. In addition, it improves the performance of the buck–boost converter by continuously applying additional

Table 1: Comparison of the modified bidirectional DC–DC converter based on the buck–boost topology

Modification technique	References	Number of components					Converter design		Maximum efficiency (%)	
		Switch	Diode	Inductor	Coupled inductor	Capacitor	Highest f_{sw} (kHz)	Maximum power rating (W)		
SR technique	[62]	4	–	1	–	3	50	200	94.1	94.8
	[63]	4	–	2	–	1	30	200	94.1	95.1
Coupled inductor	[65]	3	3	1	1	4	100	2000	96.3	95.3
	[66]	3	–	2	–	1	50	200	96.2	96.7
	[68]	2	–	–	1	2	50	400	91.3	93.3
	[69]	2	–	1	1	5	20	100	–	–
	[70]	3	7	1	2	5	10	285	–	–
DAB topology	[78]	4	–	2	1	4	100	1000	96.6	96.6
	[79]	4	–	1	1	4	100	1000	96.0	96.0
Cascaded topology	[81]	4	–	1	–	3	45	160	98.0	98.0
	[82]	8	–	2	–	3	25	1100	98.0	98.0
	[83]	8	–	2	–	3	25	1100	98.0	98.0
Interleaved topology	[86]	8	–	4	–	4	200	1500	95.2	–
	[87]	8	–	4	–	2	20	5000	–	–
	[88]	4	–	2	–	2	1000	1000	97.5	97.0
Resonant circuit	[94]	4	–	2	–	3	16	1000	96.0	94.2
	[95]	2	–	2	–	3	40	2000	97.8	98.4
	[96]	4	2	1	1	3	100	150	96.4	94.5
	[97]	4	1	4	–	1	100	150	98.2	97.5
Multi-input topology	[100]	3	5	1	1	4	830	200	94.0	92.0
	[101]	6	7	5	–	5	–	8300	–	–
	[102]	4	2	2	–	2	75	–	–	–
	[103]	3	3	1	–	2	10	200	–	–
	[104]	3	1	2	–	3	50	400	97.5	92.0
	[105]	4	5	1	1	4	50	500	94.3	95.2
Auxiliary clamp circuit	[106]	6	2	1	–	1	10	350	96.7	97.8
	[108]	4	4	2	–	1	50	200	97.5	97.7
	[109]	3	–	3	–	3	10	270	–	–
	[110]	5	–	3	–	6	100	350	95.5	95.2
	[111]	4	8	–	3	6	50	600	97.2	97.5

circuits or components. As a result, a novel topology is proposed, making many choices to be applied in the systems. In this paper, 30 studies that have proposed the novel bidirectional DC–DC converter based on the buck–boost topology are divided into eight modification techniques. For comparison

Table 2: Comparison of the applications of modified bidirectional DC–DC converters for RES power plants

Modification technique	References	Converter applications for RES-based power plants		
		PV	WT	FC
SR technique	[62]	✓		✓
	[63]	✓		✓
Coupled inductor	[65]	✓	✓	✓
	[66]	✓		✓
	[68]	✓		✓
	[69]	✓		✓
	[70]	✓	✓	✓
DAB topology	[78]	✓	✓	✓
	[79]	✓	✓	✓
Cascaded topology	[81]	✓	✓	
	[82]			✓
	[83]			✓
Interleaved topology	[86]	✓		
	[87]	✓		
	[88]	✓	✓	✓
Resonant circuit	[94]	✓		
	[95]	✓		
	[96]	✓		✓
	[97]	✓		
	[97]	✓		
Multi-input topology	[100]	✓	✓	✓
	[101]	✓	✓	✓
	[102]	✓	✓	✓
	[103]	✓	✓	✓
	[104]	✓	✓	✓
	[105]	✓	✓	✓
Auxiliary clamp circuit	[106]	✓	✓	✓
	[108]	✓	✓	✓
	[109]	✓		
	[110]	✓		
	[111]	✓	✓	

of the number of components used, the converter design includes the highest switching frequency with the maximum power tested and the maximum efficiency result, shown in Table 1.

Increasing the use of switching devices is the same as increasing the control complexity due to the fact that each of them should be determined by their trigger signal. It is dominated when bidirectional DC–DC converters in cascading and interleaved topologies are modified. However, both functions are not the same, in which the cascaded topology is prepared to obtain more gain to increase the voltage from the storage side to the load side requirement, or the input inverter in discharging/boost mode during RES-based power plants, where a PV system, WT and FC do not work. Meanwhile, the interleaved topology focuses on the suppressed ripple and decreases the filter capacitance used by ripple cancellation. Therefore, it offers good power quality for other parts connected to the proposed converter based on the interleaved topology. The biggest problems are the non-ideal factors caused by an asymmetric layout and parasitic elements in the components used. These factors can cause the phases of an interleaved converter to flow current in an unbalanced way.

Adding switch devices is used to replace the diode role for other modification techniques: SR, DAB topology and resonant circuit. So, the conduction loss can be minimized, except for the multiport/multi-input topology. A diode is still needed to determine where the primary source works in the overall system, because this configuration is acceptable for more than one renewable energy source. In the form of a magnetic component, the inductor is the primary component compared to the CI. The advantages of the inductor are that they are simpler and result in a more compact design for the proposed converter, while the CI is more appropriate for minimum conduction losses and core losses. Also, researchers usually use coupled inductors to stop leakage inductance from going around the proposed circuit.

Furthermore, the auxiliary clamp circuit is utilized to help the proposed bidirectional DC–DC converter achieve soft-switching conditions by suppressing the switching surge. Also, it is used to mitigate voltage spikes caused by recovering the leakage energy in the CI [112]. Therefore, the auxiliary clamp circuit is effectively combined with the proposed converter with the CI. Therefore, both advantages can be obtained and offer an improved efficiency result. But the attention point should be considered in terms of

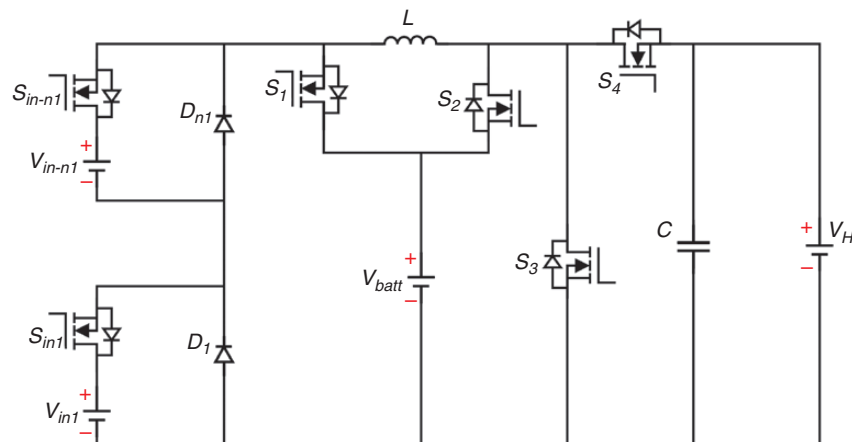


Fig. 34: Proposed converter based on multi-input topology of Varesi et al. [106].

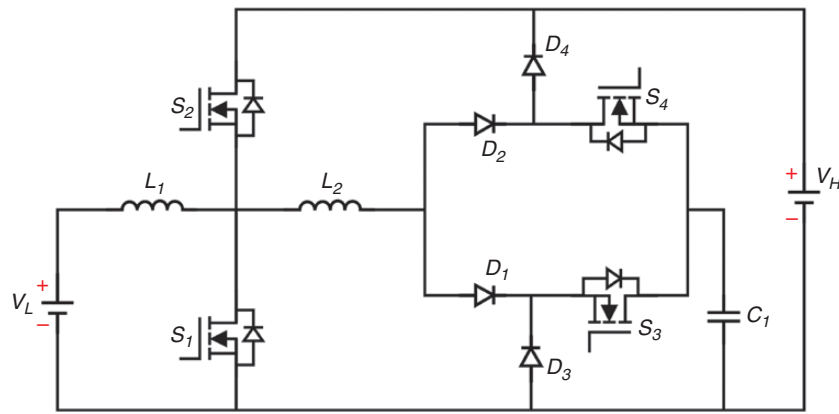


Fig. 35: Proposed converter based on multi-input topology by Yang and Do [108].

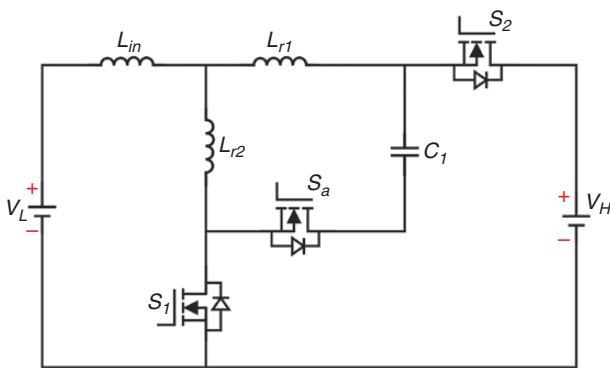


Fig. 36: Proposed converter based on the auxiliary clamp circuit [109].

its cost and the large amount of current that flows through the circuit.

The switching-frequency rate is the most important part of any DC–DC converter, including a bidirectional converter based on the buck–boost topology. Researchers should have predicted the advantages and otherwise to the level determined depending on the strategy control used, respectively. The outline is the high switching frequency used to reduce the component size and vice versa. For a detailed guide, Fang and Redl [113] have proposed determining the switching frequency step by step.

Considering each modification technique explained in this paper, the multi-stage configuration is more appropriate for the middle to high power. It consists of DAB topology, cascaded topology and interleaved topology. The SR technique, the CI, the resonant circuit and the auxiliary clamp circuit are dedicated to low-power applications. Meanwhile, the multiport/multi-input topology has high flexibility for both power-level applications. Furthermore, it is critical to support the research and development of microgrid applications [114–117] that use RES, storage systems and bidirectional DC–DC converters. Table 2 shows a comparison of each modified technique used, together with a reference, for RES-based power generation.

Regarding all the modification techniques discussed, the final comparison is shown in Fig. 39 by giving Assessments 1 to 5, focusing on six parameters to provide information for the further development of the bidirectional DC–DC converter based on the buck–boost topology. It consists of the number of

components (NC), control complexity (CC), the power-rating applications (PR), soft-switching ability (SS), the efficiency result and minimized loss ability (ML). Note that the NC and CC have reversed or negative values, which means that the best configuration results in a smaller value in the graph. Meanwhile, the PR, SS and ML have a linear trend along with an increase in the value of the graph.

4 Conclusion

This paper discussed the existence of a non-isolated bidirectional DC–DC converter modification based on the buck–boost topology. From 30 research articles published in the last 10 years, 29 proposed converters are chosen and grouped into eight modification strategies with their closest method. All the benefits and drawbacks of the modified bidirectional DC–DC converter are discussed. The NC, control complexity, power-rating applications, soft-switching ability, efficiency results and the ability to minimize losses were compared across all suggested converters. In addition, each modified bidirectional DC–DC converter has been compared from a renewable-energy-based power-generation-source perspective. The comparison results presented in this study are intended to serve as a reference for designing and constructing a new non-isolated bidirectional DC–DC converter based on the buck–boost architecture. Modification by multiport/multi-topology can be more considered to build and combine several renewable-energy power plants that are clean and will last for a long time.

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Conflict of interest statement

We have no conflicts of interest to disclose. All authors declare that they have no conflicts of interest.

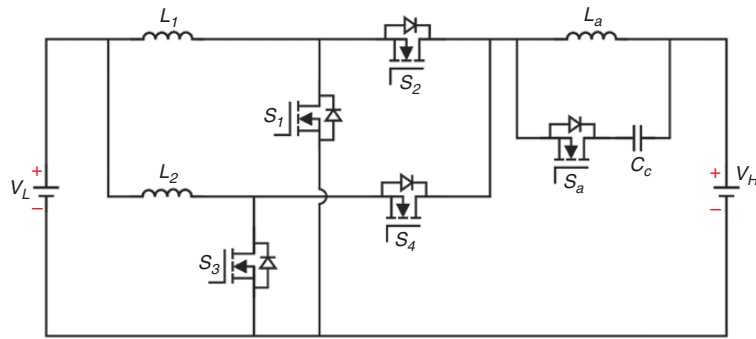


Fig. 37: Proposed converter based on auxiliary clamp circuit [110].

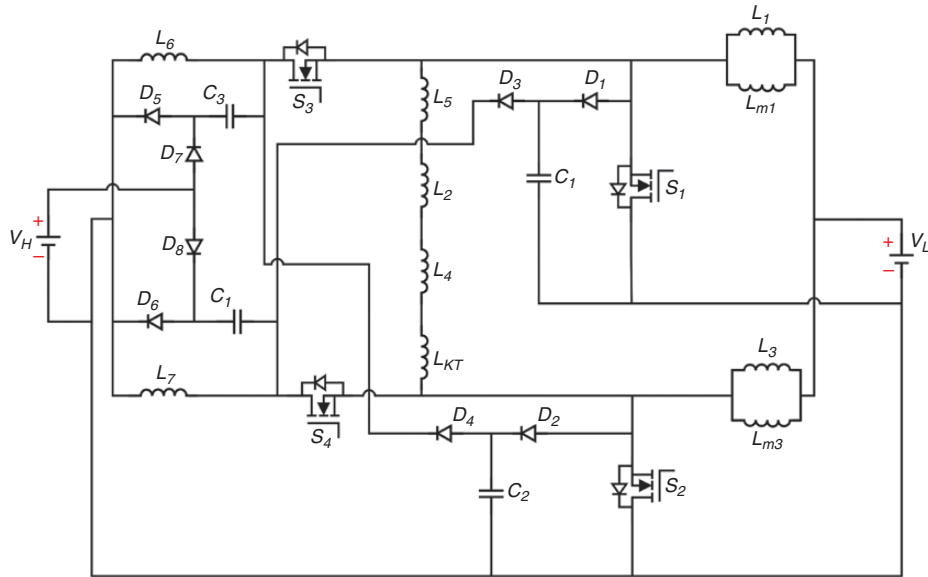


Fig. 38: Proposed converter based on the auxiliary clamp circuit [111].

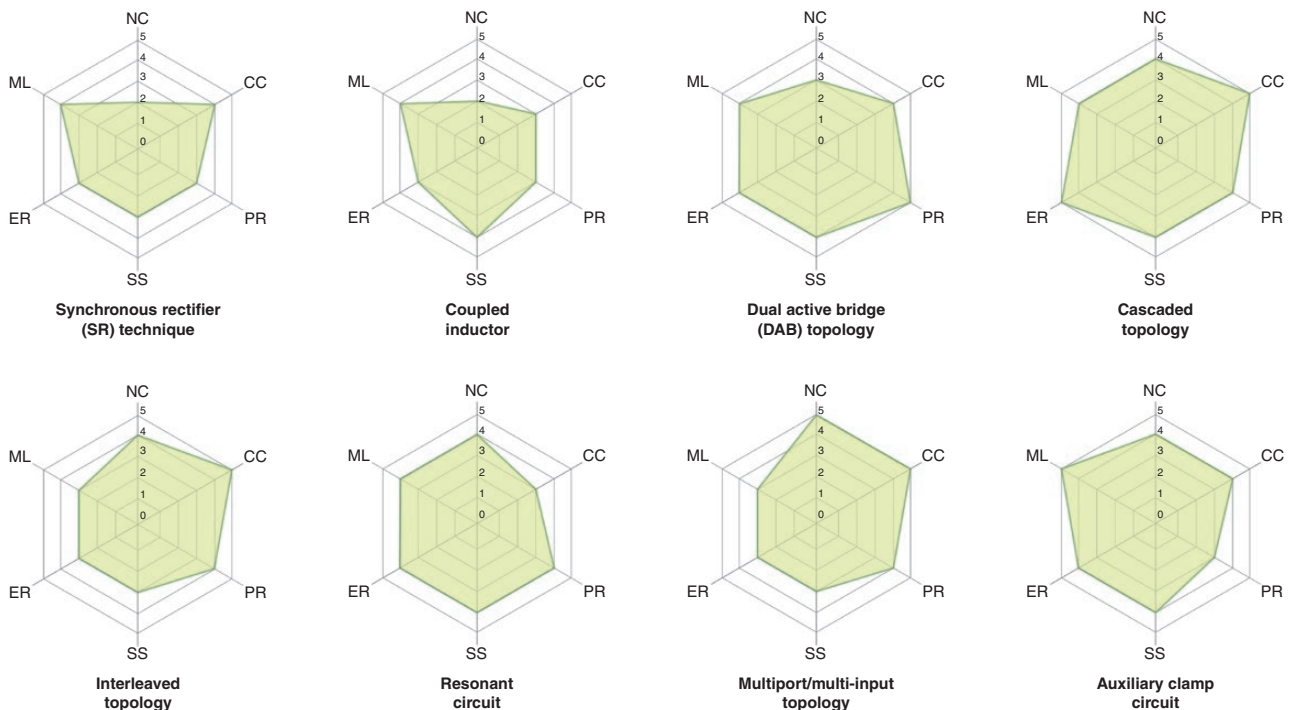


Fig. 39: Performance comparison of the modification technique for the non-isolated bidirectional DC-DC converter.

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